

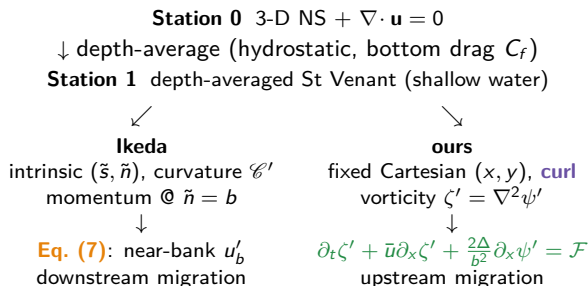
From Navier–Stokes to the Vorticity–Meander Model

Rossby Palooza — Alluvial Rivers & Jet streams

Derivation companion to `numerical/dedalus_meander/`

The map: one parent, two models

Both models start from the **same** 3-D Navier–Stokes + mass conservation and the **same** depth average. They **fork** at one step (Station 2): Ikeda evaluates the momentum balance in channel-fitted coordinates; we take the **curl** in a fixed frame.



Station 0–1: Navier–Stokes → shallow water (shared)

Station 0. Incompressible NS with free surface $z = \tilde{\xi}$, bed $z = \tilde{\eta}$:

$$\nabla \cdot \mathbf{u} = 0, \quad \frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho}\nabla p - g\hat{\mathbf{z}} + \frac{1}{\rho}\nabla \cdot \boldsymbol{\tau}.$$

Station 1. Four assumptions collapse 3-D → 2-D:

- (i) **hydrostatic** ($H \ll L$): $\partial_z p = -\rho g \Rightarrow p = \rho g(\tilde{\xi} - z)$, so horizontal $\nabla p = \rho g \nabla \tilde{\xi}$;
- (ii) depth-integrate continuity + horizontal momentum ($\tilde{\eta} \rightarrow \tilde{\xi}$);
- (iii) vertical stress → **bottom drag** $\tau_b = \rho C_f |\mathbf{U}| \mathbf{U}$ (top ≈ 0);
- (iv) **uniform vertical profile** — *this discards the secondary (helical) flow.*

$$\partial_t h + \nabla \cdot (h\mathbf{U}) = 0, \quad \partial_t \mathbf{U} + (\mathbf{U} \cdot \nabla) \mathbf{U} = -g \nabla \xi - \frac{C_f |\mathbf{U}| \mathbf{U}}{h}$$

Assumption (iv) is why Ikeda must *re-inject* the secondary flow as the closure A ; our model simply omits it (flat rigid bed).

Station 2 (ours): low Froude $\Rightarrow$ rigid lid + curl

$\text{Fr} = U_0/\sqrt{gH} \in (0.1, 0.3)$, so $\text{Fr}^2 \ll 1$. Momentum balance fixes the surface:

$$\underbrace{(\mathbf{U} \cdot \nabla) \mathbf{U}}_{\sim U_0^2/L} \sim \underbrace{g \nabla \eta}_{\sim g\eta/L} \Rightarrow \frac{\eta}{H} \sim \frac{U_0^2}{gH} = \text{Fr}^2.$$

Feed into continuity $\partial_t \eta + H \nabla \cdot \mathbf{U} + \nabla \cdot (\eta \mathbf{U}) = 0$:

$$\boxed{\nabla \cdot \mathbf{U} = O(\text{Fr}^2) \approx 0} \Rightarrow u = -\partial_y \psi', \quad v = +\partial_x \psi' \quad (\text{streamfunction}).$$

Key consistency point: Fr^2 kills the *divergence* (continuity), not the *advection*: in momentum, advection and $g \nabla \eta$ **balance at $O(1)$** (that is what selects η). The surface becomes a slaved pressure — and the **curl** of a gradient is zero:

$$\boxed{\partial_t \zeta + \mathbf{U} \cdot \nabla \zeta = \text{curl}_z \left[-\frac{C_f}{H} |\mathbf{U}| \mathbf{U} \right], \quad \zeta = \partial_x v - \partial_y u = \nabla^2 \psi}$$

This is the low-Mach $\leftrightarrow$ incompressible analogue with $\text{Fr} \leftrightarrow \text{Ma}$: divergence = $O(\text{Fr}^2)$, momentum advection kept, pressure a Lagrange multiplier.

Station 2 (Ikeda): channel coordinates + curvature

Glue coordinates to the meandering centreline: $\tilde{s}$ arclength, $\tilde{n}$ transverse, curvature $\mathcal{C}(\tilde{s})$. Walls sit at $\tilde{n} = \pm b$ *by construction*; meandering lives in the metric. St Venant becomes

$$\tilde{u}\partial_{\tilde{s}}\tilde{u} + \tilde{v}\partial_{\tilde{n}}\tilde{u} + \underbrace{\mathcal{C}\tilde{u}\tilde{v}}_{\text{curv.}} = -g\partial_{\tilde{s}}\tilde{\xi} - \frac{\tilde{\tau}_s}{\rho\tilde{h}}, \quad \tilde{u}\partial_{\tilde{s}}\tilde{v} + \tilde{v}\partial_{\tilde{n}}\tilde{v} - \underbrace{\mathcal{C}\tilde{u}^2}_{\text{centrif.}} = -g\partial_{\tilde{n}}\tilde{\xi} - \frac{\tilde{\tau}_n}{\rho\tilde{h}}$$

Expand in $\nu = b/r_0 \ll 1$:

$$O(1): C_f U^2 = gHI, \quad UH = q_w \text{ (uniform base flow);}$$

$$O(\nu): U^2 \mathcal{C}' = g \partial_{\tilde{n}} \xi' \quad (\text{superelevation balance});$$

$$O(\nu^2): U \partial_{\tilde{s}} u' = -g \partial_{\tilde{s}} \xi' - C_f \frac{U^2}{H} \left(2 \frac{u'}{U} - \frac{\xi'}{H} + \frac{\eta'}{H} \right).$$

The factor **2** in $2u'/U$ is the linearised $|\mathbf{U}|\mathbf{U}$ (streamwise drag twice the cross-stream). This is a *field* equation at all $\tilde{n}$.

Two closures. Integrate $O(\nu)$: superelevation $\xi' = \frac{1}{g}\mathcal{C}' U^2 \tilde{n}$; import the 3-D secondary-flow result $\eta'/H = -A\mathcal{C}'\tilde{n}$ ($A \approx 2.89$ alluvial). Substitute and evaluate at $\tilde{n} = b$, $u'_b = u'(\tilde{n} = b)$:

$$U \frac{\partial u'_b}{\partial \tilde{s}} + 2 \frac{U}{H} C_f u'_b = b \left[-U^2 \partial_{\tilde{s}} \mathcal{C}' + C_f \mathcal{C}' \left(\frac{U^4}{gH^2} + A \frac{U^2}{H} \right) \right] \quad (7)$$

term	origin	physics
$-U^2 b \partial_{\tilde{s}} \mathcal{C}'$	$-g \partial_{\tilde{s}} \xi'$	free-vortex (inner bank fast, no lag)
$C_f b \mathcal{C}' U^4 / gH^2$	superelevation ξ'/H	deeper outer column, $\propto \text{Fr}^2$
$C_f b \mathcal{C}' A U^2 / H$	secondary flow $-\eta'/H$	bed scour, dominant ($A/\text{Fr}^2 \approx 30$)

LHS: advection + friction relaxation over $H/2C_f$. Result: u'_b peaks $\sim 0.18\lambda$ *downstream* of the apex $\Rightarrow$ bends grow & migrate **downstream** ($\omega_0 > 0 \forall k$).

Station 3–4 (ours): base jet, β , friction closures

Base state (observed parabolic jet, Bahmanpouri 2022):

$$\bar{u}(y) = U_0 + \frac{\Delta}{b^2}(b^2 - y^2) = 1 - Dy^2, \quad \boxed{\bar{\zeta}_y = -\bar{u}_{yy} = \frac{2\Delta}{b^2} = 2D > 0}$$

the constant positive vorticity gradient — the channel's β .

Linearise $\mathbf{U} \cdot \nabla \zeta \rightarrow \bar{u} \partial_x \zeta' + v' \bar{\zeta}_y$ ($v' = \partial_x \psi'$):

$$\boxed{\partial_t \zeta' + \bar{u}(y) \partial_x \zeta' + \frac{2\Delta}{b^2} \partial_x \psi' = \mathcal{F}[\psi'], \quad \zeta' = \nabla^2 \psi'}$$

Friction $\mathcal{F} = \text{curl}_z[-\frac{C_f}{H}|\mathbf{U}|\mathbf{U}]$, two readings:

$$\mathcal{F}_{\text{ray}} = -\gamma \zeta', \quad \mathcal{F}_{\text{mom}} = -\gamma [2\bar{u} \psi'_{yy} + 2\bar{u}_y \psi'_y + \bar{u} \psi'_{xx}]$$

($\gamma = C_f b/H$.) They differ only by whether $|\mathbf{U}|\mathbf{U}$'s anisotropy is kept; this is the $E \leftrightarrow 2E$ degeneracy and the $+2\gamma$ in the dispersion.

Boundary: the erodible bank (the engine)

Rigid banks $\psi'(\pm b) = 0$: the parabolic jet has **no inflection point** $\Rightarrow$ Rayleigh-neutral. So the instability must come from the wall.

Erodible bank (deck p.7): the wall streamfunction relaxes toward the interior,

$$\partial_t \psi'|_{y=\pm b} = \frac{\varepsilon C_f U_0}{b} (\psi'|_{y=0} - \psi'|_{y=\pm b})$$

not an external forcing — the bank is driven by the flow it encloses.

Reynolds, not Taylor. On the slide the split $\psi_i = \bar{\psi}(y_i) + \hat{\psi}_i e^{i(kx - \omega t)}$ is a base+perturbation (Reynolds) decomposition. The physical bank displacement η is not a state variable; it is recovered from the wall streamfunction by the kinematic map (total ψ constant along the wall streamline):

$$\psi'(\pm b) = \bar{u}(\pm b) \eta = U_0 \eta \quad \Longleftrightarrow \quad \eta = \psi'(\pm b)/U_0$$

So seeding a sinusoidal wall streamfunction *is* seeding a sinusoidal meander; $y = \pm b$ is the fixed base wall (linear BC applied there to $O(\text{amp}^2)$).

The three extra terms (vs Ikeda), from the curl

Ikeda (7) is his momentum pair curled and read at $\tilde{n} = b$; with $\zeta'_l \equiv -\partial_{\tilde{n}} u' = -u'_b/b$,

$$U\partial_{\tilde{s}}\zeta'_l + 2C_f\frac{U}{H}\zeta'_l = U^2\partial_{\tilde{s}}\mathcal{C}' - C_f\mathcal{C}'\frac{U^4}{gH^2} - C_f\mathcal{C}'\frac{AU^2}{H}.$$

Against ours $\partial_t\zeta' + \bar{u}\partial_x\zeta' + 2D\partial_x\psi' = \mathcal{F}$, $\zeta' = \psi'_{yy} - k^2\psi'$:

extra term	born at	Ikeda lacks it because
$\partial_t\zeta'$	Station 2 (kept ∂_t)	quasi-steady (flow slaved to $\mathcal{C}'$)
$2D\partial_x\psi' = \nu'\bar{\zeta}_y$	Station 4 (linearise)	plug flow $\bar{\zeta} \equiv 0$: <i>identically zero</i>
$-k^2\psi'$ in ζ'	Station 2 (vorticity)	keeps only shear $-\partial_{\tilde{n}}u'$ (parabolic in $\tilde{s}$)

Degenerate check: set $D = 0$, $\partial_t = 0$, drop $-k^2\psi' \Rightarrow$ our operator collapses to $U\partial_{\tilde{s}} + 2C_f U/H$; add back sources $\Rightarrow$ Eq. (7). **Same parent, different terms kept.** These three turn “passive response to curvature” into “a wave with its own retrograde phase speed.”

The 3-level closure: where friction & erosion live

Levels $\psi_1(+b), \psi_2(0), \psi_3(-b)$, sinuous $\hat{\psi}_1 = \hat{\psi}_3$; 3-point Laplacian $\hat{\zeta}_2 = 2\hat{\psi}_1 - (2 + k^{*2})\hat{\psi}_2$.

Friction lives in the **interior (centre) vorticity** equation:

$$(-i\omega^* + ik^*)\hat{\zeta}_2 + \underbrace{2iDk^*\hat{\psi}_2}_{\beta} = \underbrace{\hat{\mathcal{F}}_2}_{\text{friction}}$$

Erosion lives in the **bank** equation (dynamic BC):

$$(-i\omega^* + E)\hat{\psi}_1 = E\hat{\psi}_2, \quad E = \varepsilon C_f U_0 / (U_0 + \Delta)$$

Two equations, two unknowns $\Rightarrow \det M(\omega^*) = 0$. Feedback loop: centre feels the banks (the $2\hat{\psi}_1$ in $\hat{\zeta}_2$); banks erode toward the centre. **No erosion (rigid) $\Rightarrow$ no instability.**

“Friction at the centre” is the $N=3$ artefact (one interior node); in the continuum it acts at *all* interior y . Ikeda’s friction is likewise a bulk term — Eq. (7) merely reads it at $\tilde{n} = b$.

The Dedalus solver = the continuum version

The 3-level closure is the crudest $N = 3$ discretisation. `dedalus_meander` resolves the full y -profile (Chebyshev), **5 fields / 5 equations**:

unknown	equation
$\psi'(x, y)$ (Fourier $\times$ Chebyshev)	interior vorticity PDE + 2 taus
$\psi b_{\text{top}}(x)$, $\psi b_{\text{bot}}(x)$	$\psi'(y = \pm 1) = \psi b_{\pm}$ (Dirichlet)
$\tau_1(x)$, $\tau_2(x)$ (numerical)	$\partial_t \psi b_{\pm} = E(\psi'(y=0) - \psi b_{\pm})$

Physical d.o.f.: ψ' + two banks; taus are the spectral BC device (2, since ∇^2 is 2nd order in y).

Verification. The Dedalus EVP reproduces `vorticity_lib`'s N -point GEP eigenvalue to 10^{-10} ; IVP-measured (σ, c) match to 0.01%; rigid banks neutral to 10^{-8} ; the varicose mode decays at exactly $-iE$.

Bank mode (verified, $D = 0.6, \gamma = 0.05$, rayleigh):

- **Growth band** $k^{*2} < 2D$ (interior amplified iff $|\hat{\psi}_2| > |\hat{\psi}_1|$), peak $k^* \approx 0.44$.
- **Phase speed** $c^* = \text{Re} \omega^* / k^* < 0$ for *all* k^* : the bends migrate **upstream**, $c^* \rightarrow -ED/\gamma$ as $k^* \rightarrow 0$.
- Contrast Ikeda: $\omega_0 > 0 \forall k$, always **downstream**.

Momentum flux (deck p.6): with friction the wave tilts and

$$\partial_y \overline{u'v'} = \frac{\gamma}{2D} \overline{\zeta_2'^2} > 0$$

carries streamwise momentum from the jet core *transversely* to the banks (zero for the steady forced wave at $\gamma = 0$).

Open question (codified): the deck's growth-rate *peaks* sit $2.3\text{--}4.4\times$ above the literal model; the momentum closure closes $\sim 1/3$ of the gap — friction closure alone does not explain it (suspects: wavy-bank pressure drag; growth-axis normalisation).

New prediction: the group-velocity sign flip

Phase and group velocity **disagree** — the Rossby signature. For $D = 0.6, \gamma = 0.05$:

wavelength	k^*	crests c^*	momentum $c_g = d\omega_r/dk$
very long ($\lambda/2b > 10.7$)	< 0.29	$-x$ (up)	$-x$ upstream
growth-band body	$0.29-1.1$	$-x$ (up)	$+x$ downstream (max $+0.13$ at $k^* \approx 0.7$)
short (decaying)	> 1.1	≈ 0	$+x, \rightarrow 0$

crests retrograde (upstream) for all λ ; momentum/energy flips sign at $k^* \approx 0.29$ ($\lambda/2b \approx 10.7$)

Only waves longer than ~ 11 channel widths carry momentum upstream; everything shorter carries it downstream while its crests still crawl upstream. Testable with a localized wave-packet IC (track the envelope vs the crests).

Why the movies look “advective,” not “erosive”

The bank mode both **grows** (σ) and **propagates** (c^*):

$$\psi b(x, t) = |a| e^{\sigma t} \cos(k(x - c^* t) + \varphi).$$

- For long waves $|c^*|$ is large (-0.6) while σ is small (0.055): the **translation dominates the eye** $\Rightarrow$ “looks like advection.”
- The field panels are amplitude-*normalized* (fixed display size), so the growth is hidden — it lives in the gain counter / the $\log |a|$ curve.
- It is **not** mean-flow advection: that would be $+x$ at speed ~ 1 ; this is $-x$ at 0.6 . The bank equation has *no* ∂_x term — the motion is the eigenmode’s retrograde phase speed inherited through relaxation.

The 4-panel movies fix the perception: σ (growth) and c^* (crests) vs c_g (momentum) are shown separately, and the $\log |a(t)|$ curve makes the erosion growth explicit.

- One parent: 3-D NS + mass $\rightarrow$ depth-averaged shallow water.
- **Fork at Station 2**: Ikeda evaluates momentum in intrinsic coordinates ($\rightarrow$ Eq. 7 at $\tilde{n} = b$); we take the **curl** in a fixed frame ($\rightarrow$ the vorticity equation).
- Three curl-born terms ($\partial_t \zeta'$, the β term $2D\partial_x \psi'$, the $-k^2 \psi'$ share) turn Ikeda's passive, always-downstream response into a **channel Rossby wave** that migrates **upstream**.
- Friction is a bulk interior term (both models); erosion is a dynamic boundary condition (the engine; rigid $\Rightarrow$ neutral).
- `dedalus_meander` = the continuum solver (5 fields / 5 eqs), verified against the theory GEP to 10^{-10} .
- Prediction: **group-velocity sign flip** at $k^* \approx 0.29$ — crests upstream everywhere, momentum upstream only for $\lambda/2b > 10.7$.